Rà soát từng câu — `DTAP Finally v2/main_document.tex`

Mục tiêu: bài dễ đọc, văn phong phù hợp một bài báo toán tối ưu, mỗi câu chặt chẽ, tường minh, ngắn gọn. **Chưa sửa gì trong bài**; bảng này chỉ ghi kết quả rà soát.

Tiêu chí

Một câu đạt yêu cầu khi người đọc chỉ hiểu được **một nghĩa**, hiểu được **mà không phải đoán**, nội dung **đúng như toán học đã chứng minh**, câu **nối được** với câu trước và sau, và **không có chữ thừa**. Mỗi lỗi được gán một hoặc nhiều mã:

Mã	Loại	Dấu hiệu
A	Mơ hồ (đọc ra hai nghĩa)	Đại từ hay chủ ngữ không rõ; phạm vi lượng từ không rõ; bổ ngữ treo; câu dễ đọc nhầm cấu trúc.
B	Tối nghĩa (một nghĩa nhưng phải đoán)	Thuật ngữ, ký hiệu dùng trước khi định nghĩa; biệt ngữ, ẩn dụ; chồng danh từ; nhắc tới quy trình hay tên gọi chưa giới thiệu.
C	Thiếu chặt chẽ	Khẳng định rộng hơn điều đã chứng minh; thiếu giả thiết hay lượng từ; một ký hiệu hai nghĩa; một khái niệm nhiều tên.
D	Liên kết	Câu hoặc đoạn không nối với phần trước; từ nối sai logic; một ý lặp ở nhiều nơi; nội dung đặt sai chỗ.
E	Câu dài hoặc nặng	Trên 30 từ, hoặc từ 3 mệnh đề trở lên, hoặc nhiều phủ định chồng nhau; hoặc chuỗi câu quá ngắn rời rạc.
F	Thừa / lan man	Phép thử xóa: bỏ câu mà không mất thông tin thì câu thừa; câu lệch mục đích của đoạn, lặp điều đã nói, hoặc rào đón.

Súc tích nghĩa là mỗi câu nói một ý cần thiết, chính xác, bằng ít chữ nhất mà người đọc vẫn không phải đoán. Nén quá mức thành biệt ngữ là lỗi B, không phải súc tích.

Mức độ: **[1] phải sửa** (sai hoặc có thể làm phản biện nghi ngờ kết quả), **[2] nên sửa** (gây khó hiểu), **[3] tùy chọn** (văn phong).

Kết quả tổng quát

- Đã rà **503 câu/đơn vị** (gồm tiêu đề, chú thích bảng, các bước của Thuật toán 1; không gồm phần thân bảng số liệu và danh mục tài liệu tham khảo).
- **452 câu không cần sửa.**
- **51 câu có vướng lỗi:** 1 câu mức [1], 22 câu mức [2], 28 câu mức [3].
- Theo loại (một câu có thể có nhiều mã): A mơ hồ: 5, B tối nghĩa: 13, C thiếu chặt chẽ: 6, D liên kết: 3, E câu dài/nặng: 21, F thừa/lan man: 3.
- Câu trên 30 từ: 31; trong đó 11 câu được đề nghị tách (mức [2]). Các câu dài còn lại là danh sách tham số hoặc chứa công thức, có thể giữ.
- Phần chứng minh (Mục 3.6–3.9) đạt yêu cầu: các bước đều có lý do, lượng từ rõ, không có câu thừa.
- Hai câu cần lưu ý nhất: câu về công thức (8) ”whose maximizers…” (mức [1], đọc ra nghĩa sai) và câu Introduction ”the proof uses this weight only to…” (mức [2], chưa chính xác; câu này đang được letter R1.1 trích nguyên văn).

Vấn đề xuyên suốt cả bài

Mã	Mức	Vấn đề và cách sửa
C	[2]	Tên của J không thống nhất: violation measure, merit function, error function, feasibility residual, full residual. Chọn một tên (ví dụ ”residual function”) cho J; dành ”violation” cho v_i.
C	[3]	Tên của bước trên track lồi không thống nhất: Polyak step, Polyak-type step, residual-scaled step. Dùng ”residual-scaled Polyak-type step”.
C	[3]	Ký hiệu trùng nghĩa ở Mục 4: η (tốc độ học AFRSFCV và độ dài bước DTAP), T (số bước AFRSFCV, thời điểm cuối PNN, và T_k), t (chỉ số lớp, thời gian, và t = $\rho^T w$ trong chứng minh Mệnh đề 4.1).
B	[2]	Hai tên không được định nghĩa: ”Tchebyshev scalarization” (tiêu đề Mục 3.1, từ khóa) và ”solver diagnostic” (Mục 4 mở đầu, chú thích Bảng 1).
D	[3]	Tính lồi của Ω được nói ba lần (Introduction, Mục 2, Mục 3.3); thông tin quy tắc chọn ngưỡng nói hai lần (Mục 4.1.3 và 4.2.2).
E	[2]	Nhiều câu dài trên 30 từ, tập trung ở Mục 4.4 và Mục 4.2 (xem số liệu ở Kết quả tổng quát). Các câu nên tách được đánh dấu E [2] trong bảng; các câu dài còn lại là danh sách tham số hoặc chứa công thức, có thể giữ.

Bảng rà soát từng câu

Cột 1: câu nguyên văn (số trong ngoặc vuông là thứ tự câu; [công thức (n)] là công thức hiển thị). Cột 2: có lỗi hay không, mã và mức. Cột 3: “Giữ nguyên” hoặc cách sửa cụ thể. Hàng tô hồng là câu có lỗi.

Câu	**Có lỗi không?**	**Không cần sửa / Cách sửa**
[1] We study a nonlinear split feasibility problem whose output set is a normal box and whose component functions are differentiable and each convex or pseudoconvex.	Không	Giữ nguyên
[2] We propose the Dual-Track Adaptive Projection (DTAP) algorithm, which splits the constraints into a convex and a pseudoconvex track and works on the nearly most violated constraints of each.	**Có** — B [3]	Cụm ”works on” không nói rõ làm gì với các ràng buộc. Có thể viết: ”...and, on each track, uses the gradients of the nearly most violated constraints.”
[3] DTAP combines a residual-scaled Polyak-type step on the convex track with a safeguarded diminishing step on the pseudoconvex track through a residual-responsive proportional weight, and then projects onto the base convex set.	**Có** — E [3]	32 từ nhưng chỉ một ý; có thể giữ. Nếu muốn ngắn: ”DTAP combines a residual-scaled Polyak-type step on the convex track with a safeguarded diminishing step on the pseudoconvex track by a residual-responsive proportional weight, and projects onto the base convex set.”
[4] Its step sizes require no prescribed operator norm, Lipschitz constant, or Hölder modulus; the proof uses local bounds on the bounded iterate region.	Không	Giữ nguyên
[5] Under consistency and continuous differentiability, we prove that the whole sequence converges strongly to a feasible point.	Không	Giữ nguyên
[6] As an application, we formulate feature selection as such a problem, with constraints built from fractional and fuzzy-rough feature-selection criteria.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[7] Experiments on UCI datasets show that the selected features keep the classification accuracy close to that of the full feature set, and test problems with a nonconvex pseudoconvex constraint exercise the pseudoconvex track; we also compare DTAP with subgradient methods and study its sensitivity to the starting point and the step scale.	**Có** — E [2]	52 từ, hai mệnh đề nối bằng dấu chấm phẩy. Tách hai câu: ”Experiments on UCI datasets show that the selected features keep the classification accuracy close to that of the full feature set. Test problems with a nonconvex pseudoconvex constraint exercise the pseudoconvex track; we also compare DTAP with subgradient methods and study its sensitivity to the starting point and the step scale.”
[8] Split feasibility problem; Pseudoconvex optimization; Tchebyshev scalarization; Adaptive step size; Strong convergence; Feature selection 47J25; 90C25; 90C26; 65K05; 68T05	Không	Giữ nguyên
[9] **Introduction**	Không	Giữ nguyên
[10] The split feasibility problem (SFP) was introduced by Censor and Elfving [4] as a model for constrained inverse problems.	Không	Giữ nguyên
[11] Given nonempty closed convex sets $C \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ and a matrix $\mathcal{A} \in \mathbb{R}^{m\times n}$, it asks to [công thức (1)]	Không	Giữ nguyên
[12] Byrne’s CQ algorithm [3] iterates [công thức (2)]	Không	Giữ nguyên
[13] Here $\|\mathcal{A}\|$ is the spectral norm of $\mathcal{A}$.	Không	Giữ nguyên
[14] Relaxed variants project onto supporting half-spaces [17]; self-adaptive steps avoid estimating $\|\mathcal{A}\|$ [20, 22, 10]; operator splitting [21] and alternating direction methods [18] decompose the problem.	Không	Giữ nguyên
[15] These ideas are standard; DTAP adapts them to the nonlinear case.	**Có** — A [2]	”them” gồm cả biến thể chiếu lên nửa không gian tựa, mà DTAP không dùng. Viết rõ: ”These ideas are standard; DTAP adapts the self-adaptive step and the decomposition to the nonlinear case.”
[16] Replacing $\mathcal{A}$ by a nonlinear map F gives the nonlinear SFP [9, 19] [công thức (3)] where $K \subseteq \mathbb{R}^n$ is nonempty, closed and convex, $Q = \prod_{i=1}^{m}(-\infty, b_i]$ is a normal box, and each F_i is differentiable and convex or pseudoconvex on K.	Không	Giữ nguyên
[17] Since pseudoconvex functions have convex sublevel sets, the feasible set $\Omega = \{x \in K : F(x) \in Q\}$ is convex.	Không	Giữ nguyên
[18] We seek a point of Ω, not a minimizer of any F_i.	Không	Giữ nguyên
[19] Because Q is a box, the ℓ_∞-distance from $F(x)$ to Q has the closed form [công thức]	Không	Giữ nguyên
[20] Li et al. [9] assume a convex aggregate proximity function, which componentwise pseudoconvexity does not ensure.	Không	Giữ nguyên
[21] Yen et al. [19] use the same max-type violation measure and active-set device, but require quasilinear components, a class incomparable with ours.	**Có** — B [3]	”active-set device” là biệt ngữ. Có thể viết: ”…use the same max-type violation measure and an active set of nearly most violated constraints, but require…”

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[22] Hu et al. [7] treat quasiconvex inequalities, which contain our class, without using the gradient inequality of convex components.	**Có** — A [3]	”our class” chưa được gọi tên. Viết: ”…treat quasiconvex inequalities, a class that contains convex and pseudoconvex functions, without using…”
[23] Convex and pseudoconvex constraints call for different steps.	Không	Giữ nguyên
[24] For a convex constraint, the gradient inequality bounds $\langle \nabla F_i(x),\, x - x^\star \rangle$, for every $x^\star \in \Omega$, from below by the violation, which justifies a residual-scaled Polyak-type step [14, 10]; for a pseudoconvex constraint, only the sign of this inner product is known, so a safeguarded diminishing step is used.	**Có** — E [2]	45 từ. Tách tại dấu chấm phẩy thành hai câu: ”For a convex constraint, the gradient inequality bounds ⟨∇F_i(x),x−x*⟩, for every x*∈Ω, from below by the violation, which justifies a residual-scaled Polyak-type step [14, 10]. For a pseudoconvex constraint, only the sign of this inner product is known, so a safeguarded diminishing step is used.”
[25] DTAP takes the two steps on separate tracks and combines them by a proportional weight; the proof uses this weight only to keep a persistently violated track active (Remark 1).	**Có** — C [2]	”uses this weight only to keep a persistently violated track active” chưa chính xác: Nhận xét 1 dùng hai tính chất, (27) là cận dưới μ_k ≥ J^cvx_k/(2V) cho track lồi, và Khẳng định 2 cho 1−μ_k ≥ 1/2 khi track giả lồi bị vi phạm kéo dài. Viết: ”…the proof uses only two properties of this weight (Remark 1): it is bounded below by a multiple of the convex-track violation, and it gives a persistently violated pseudoconvex track a weight of at least 1/2.” LƯU Ý: câu này đang được letter (R1.1) trích nguyên văn; nếu sửa phải sửa letter theo.
[26] The contributions are as follows.	Không	Giữ nguyên
[27] A dual-track method that uses the gradient inequality on the convex constraints and only sign information on the pseudoconvex ones, without assuming a convex proximity function, unlike [9], or a Hölder modulus, unlike the dynamic step of [7].	**Có** — E [3]	38 từ, hai cụm ”unlike […]” chen giữa làm khó đọc. Có thể viết: ”A dual-track method that uses the gradient inequality on the convex constraints and only sign information on the pseudoconvex ones; unlike [9], it assumes no convex proximity function, and unlike the dynamic step of [7], it needs no Hölder modulus.”
[28] Convergence of the whole sequence to a feasible point under Assumption 1 (Theorem 3.12).	Không	Giữ nguyên
[29] A feature-selection application built from fractional and fuzzy-rough criteria, whose constraints are convex (Proposition 4.1), and test problems with a nonconvex pseudoconvex constraint; the experiments report constraint activity, compare DTAP with tuned subgradient methods, and measure its sensitivity to the starting point and the step scale.	**Có** — E [2]	46 từ. Tách: ”A feature-selection application built from fractional and fuzzy-rough criteria, whose constraints are convex (Proposition 4.1), and test problems with a nonconvex pseudoconvex constraint. On both, the experiments report constraint activity, compare DTAP with tuned subgradient methods, and measure its sensitivity to the starting point and the step scale.”
[30] Section 2 gives preliminaries.	**Có** — D [3]	Ba câu rất ngắn liền nhau (mục lục). Có thể gộp: ”Section 2 gives preliminaries, Section 3 presents DTAP and its convergence analysis, Section 4 reports experiments, and Section 5 concludes.”
[31] Section 3 presents DTAP and its convergence analysis.	Không	Giữ nguyên
[32] Section 4 reports experiments, and Section 5 concludes.	Không	Giữ nguyên
[33] **Preliminaries**	Không	Giữ nguyên
[34] Throughout, $H = \mathbb{R}^n$ is the Euclidean space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[35] All convergence statements concern the norm topology.	**Có** — F [3]	Trong $\mathbb{R}^n$ mọi tô-pô chuẩn trùng nhau nên câu này hiển nhiên. Vì bài giữ cụm ”strong convergence”, nên dùng câu này để giải thích cụm đó: ”In $\mathbb{R}^n$, strong and weak convergence coincide; we keep the term strong convergence for consistency with the SFP literature.”
[36] Bounded sequences have convergent subsequences.	**Có** — F [3]	Điều hiển nhiên (Bolzano–Weierstrass). Có thể bỏ, hoặc ghi ”(Bolzano–Weierstrass)” ở chỗ dùng trong chứng minh Định lý 3.12.
[37] The standard projection and generalized convexity facts below can be found in [1, 12].	Không	Giữ nguyên
[38] For a nonempty closed convex set $K \subseteq H$, the metric projection $P_K(x)$ is the unique minimizer of $\|y - x\|$ over $y \in K$.	Không	Giữ nguyên
[39] For $x \in H$ and $p \in K$, $p = P_K(x)$ if and only if [công thức (4)]	Không	Giữ nguyên
[40] This characterization yields Lemma 3.2 below.	Không	Giữ nguyên
[41] A function $f : K \to \mathbb{R}$ is convex if $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for $x, y \in K$ and $t \in [0, 1]$.	**Có** — C [3]	Lượng từ chưa rõ: ”for x,y∈K” → ”for all x,y∈K and all t∈[0,1]”.
[42] The following weaker derivative condition was introduced by Mangasarian [11].	Không	Giữ nguyên
[43] A function f that is differentiable on an open neighbourhood of K is pseudoconvex on K if, for all $x, y \in K$, [công thức (5)]	Không	Giữ nguyên
[44] Every differentiable convex function is pseudoconvex.	Không	Giữ nguyên
[45] Pseudoconvex functions have convex sublevel sets, although the functions themselves need not be convex.	**Có** — D [3]	Tính lồi của Ω được nói ba lần (Introduction, câu này, và sau Giả thiết 1). Có thể bỏ vế này ở đây và giữ lập luận đầy đủ ở Mục 3.3.
[46] **The DTAP algorithm and its convergence analysis**	Không	Giữ nguyên
[47] We formulate the normal-box NSFP, state DTAP, and prove convergence of the whole sequence to a feasible point.	Không	Giữ nguyên
[48] **Problem statement and Tchebyshev scalarization**	**Có** — B [2]	Thuật ngữ ”Tchebyshev scalarization” xuất hiện ở tiêu đề (và từ khóa) nhưng không câu nào trong mục giải thích. Thêm vào câu về công thức (8): ”...(8) is the Tchebyshev (max-type) scalarization of the residuals (F_i(x)−b_i)^+.”
[49] Let $K \subseteq H$ be nonempty closed convex, and let $U \subseteq H$ be an open neighbourhood of K.	Không	Giữ nguyên
[50] Fix an integer $m \geq 1$, let $F = (F_1, \ldots, F_m) : U \to \mathbb{R}^m$, let $b = (b_1, \ldots, b_m) \in \mathbb{R}^m$, and set the normal box $Q := (-\infty, b_1] \times \cdots \times (-\infty, b_m]$.	Không	Giữ nguyên
[51] We study the instance of (3) [công thức (6)] and call Ω the feasible set.	Không	Giữ nguyên
[52] To turn $F(x) \in Q$ into a tractable merit function, we measure the residual in the sup-norm.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[53] For $z = F(x)$, $d_\infty(z, Q) = \inf_{q \le b} \max_{1 \le i \le m} \lvert z_i - q_i \rvert$.	Không	Giữ nguyên
[54] By the box structure the infimum separates over coordinates: if $z_i \le b_i$ then $\inf_{q_i \le b_i} \lvert z_i - q_i \rvert = 0$ (at $q_i = z_i$), and if $z_i > b_i$ then it equals $z_i - b_i$ (at $q_i = b_i$).	**Có** — C [3]	”separates over coordinates” chưa nói rõ vì sao inf của max bằng max của các inf. Viết: ”Since Q is a product of intervals, the infimum of the maximum equals the maximum of the coordinatewise infima: if z_i ≤ b_i ... ”
[55] Hence $\inf_{q_i \le b_i} \lvert z_i - q_i \rvert = (z_i - b_i)^+$ with $(t)^+ := \max\{t, 0\}$, so [công thức (7)]	Không	Giữ nguyên
[56] Define the error function [công thức (8)]	**Có** — C [2]	Một đại lượng J nhưng có nhiều tên trong bài: ”violation measure” (Intro), ”merit function” (câu trước), ”error function” (đây), ”feasibility residual” (câu sau), ”full residual” (Nhận xét 5, Kết luận). Chọn một tên, ví dụ ”residual function”, và dùng thống nhất; ”violation” để dành cho v_i.
[57] Then $x \in K$ is feasible if and only if $J(x) = 0$; when $\Omega \neq \emptyset$, the feasible points are the minimizers of J over K.	Không	Giữ nguyên
[58] Formula (8) gives a closed-form feasibility residual whose maximizers identify the most violated constraints.	**Có** — A [1]	”whose maximizers” đọc như các điểm cực đại của J (là điểm x), trong khi ý là các chỉ số i đạt max trong (8). Viết: ”In (8), the indices attaining the maximum are the most violated constraints.” (có thể gộp với câu về Tchebyshev ở trên).
[59] Section 3.2 introduces an approximate active set to handle the nonsmooth maximum.	Không	Giữ nguyên
[60] **Decomposition into two tracks**	Không	Giữ nguyên
[61] We split $I := \{1, \dots, m\}$ into a convex track I^{cvx} and a pseudoconvex track I^{ps}, each with its own step rule.	Không	Giữ nguyên
[62] The convex track uses a Polyak step [14]; the pseudoconvex track uses a safeguarded diminishing step.	**Có** — C [3]	Cùng một bước nhưng có ba tên: ”Polyak step” (đây), ”Polyak-type step” (Abstract, Intro), ”residual-scaled step”. Dùng thống nhất ”residual-scaled Polyak-type step”.
[63] A proportional weight combines the two updates.	Không	Giữ nguyên
[64] Fix a parameter $\delta > 0$.	**Có** — D [3]	δ được cố định trước khi cần tới (dùng ở (12)–(13)). Có thể chuyển câu này xuống ngay trước câu định nghĩa tập hoạt động xấp xỉ.
[65] For $x \in U$ define the componentwise violations [công thức (9)] so that $J(x) = \max_i v_i(x)$ by (8).	Không	Giữ nguyên
[66] Assumption 1 makes the partition precise.	Không	Giữ nguyên
[67] The per-track maximal violations are [công thức (10, 11)] so $J(x) = \max\{J^{\text{cvx}}(x), J^{\text{ps}}(x)\}$.	Không	Giữ nguyên
[68] To cope with the nonsmoothness of the maximum, we replace the set of indices at which the maximum is attained by a δ-enlarged approximate active set on each track (‘active’ meaning within δ of the largest violation): [công thức (12, 13)]	**Có** — E [2]	41 từ, có ngoặc giải thích chen giữa. Viết: ”To handle the nonsmooth maximum, we replace the set of maximizing indices by a δ-enlarged approximate active set on each track, which contains the indices whose violation is within δ of the largest one:”

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[69] On these sets we form the per-track local functions and directions [công thức (14, 15)]	**Có** — B [3]	”local” không có nghĩa rõ (Φ và g không phải hàm địa phương). Viết: ”On these sets we form the per-track functions and search directions”.
[70] The vectors $g^{\mathrm{cvx}}(x), g^{\mathrm{ps}}(x)$ are the two search directions, and $\Phi^{\mathrm{cvx}}(x)$ supplies the Polyak numerator on the convex track.	Không	Giữ nguyên
[71] Along the iterates we abbreviate $J_k^{\mathrm{cvx}} := J^{\mathrm{cvx}}(x^k)$, $g_k^{\mathrm{cvx}} := g^{\mathrm{cvx}}(x^k)$, $\Phi_k^{\mathrm{cvx}} := \Phi^{\mathrm{cvx}}(x^k)$, and likewise on the pseudoconvex track.	Không	Giữ nguyên
[72] **Assumptions**	Không	Giữ nguyên
[73] We assume (A1)–(A3):	Không	Giữ nguyên
[74] (a) $\Omega \neq \emptyset$ (consistency).	Không	Giữ nguyên
[75] (b) $I = \{1, \dots, m\}$ is partitioned as $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$; each $F_i : U \to \mathbb{R}$ is convex on K if $i \in I^{\mathrm{cvx}}$ and pseudoconvex on K if $i \in I^{\mathrm{ps}}$.	Không	Giữ nguyên
[76] Each $F_i : U \to \mathbb{R}$ is continuously differentiable on U.	Không	Giữ nguyên
[77] (i) $\delta > 0$ fixed.	Không	Giữ nguyên
[78] (ii) $\rho_{\mathrm{cvx}} \in (0, 2)$.	Không	Giữ nguyên
[79] (iii) $\{\gamma_k\} \subset (0, \infty)$ with $\sum_{k=0}^{\infty} \gamma_k^2 < \infty$ and $\sum_{k=0}^{\infty} \gamma_k = \infty$.	Không	Giữ nguyên
[80] The open-neighbourhood formulation permits fractional functions whose denominators are positive on K; the algorithm evaluates the functions and gradients only at points of K.	Không	Giữ nguyên
[81] Although some component functions may be nonconvex, Ω is convex: differentiable pseudoconvex functions are quasiconvex on K, so all constraint sublevel sets in K are convex.	Không	Giữ nguyên
[82] The sequence $\gamma_k := 1/(k+1)$ satisfies (A3)(iii), since $\sum_{k\geq 0}(k+1)^{-2} < \infty$ while $\sum_{k\geq 0}(k+1)^{-1} = \infty$.	Không	Giữ nguyên
[83] More generally $\gamma_k = (k+c)^{-s}$ with $c > 0$ and $s \in (1/2, 1]$ qualifies; the experiments use $s = 0.55$.	Không	Giữ nguyên
[84] **The DTAP algorithm**	Không	Giữ nguyên
[85] The Dual-Track Adaptive Projection algorithm combines the two track-specific step sizes through the proportional weight μ_k and projects the combined update onto the base convex set K.	**Có** — F [3]	Lặp lại ý đã nói ở Mục 3.2 và Abstract. Có thể gộp với câu sau: ”Algorithm 1 states DTAP; the proportional weight μ_k combines the two track-specific step sizes.”
[86] It is stated as Algorithm 1.	Không	Giữ nguyên
[87] Dual-Track Adaptive Projection (DTAP) for the normal-box NSFP (6).	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[88] Input: map $F = (F_1, \ldots, F_m)$, thresholds b, base set K, index split $I^{\text{cvx}} \cup I^{\text{ps}}$.	Không	Giữ nguyên
[89] Parameters: $\delta > 0$, tolerance $\varepsilon > 0$, $\rho_{\text{cvx}} \in (0, 2)$, sequence $\{\gamma_k\}$ satisfying (A3)(iii).	Không	Giữ nguyên
[90] Initialization: choose $x^0 \in K$; set $k \leftarrow 0$.	Không	Giữ nguyên
[91] Step 1 (per-track quantities).	Không	Giữ nguyên
[92] If $J(x^k) \le \varepsilon$, stop and return x^k; otherwise:	Không	Giữ nguyên
[93] Compute $v_i(x^k)$ for all i, and $J_k^{\text{cvx}} = \max_{i \in I^{\text{cvx}}} v_i(x^k)$, $J_k^{\text{ps}} = \max_{j \in I^{\text{ps}}} v_j(x^k)$ (each 0 if its track is empty).	Không	Giữ nguyên
[94] Form the approximate active sets $K_k^{\text{cvx}}, K_k^{\text{ps}}$ via (12)–(13).	Không	Giữ nguyên
[95] Form Φ_k^{cvx}, g_k^{cvx}, g_k^{ps} via (14)–(15).	Không	Giữ nguyên
[96] Per-track step sizes (Polyak on the convex track, safeguarded on the pseudoconvex track): [công thức]	Không	Giữ nguyên
[97] Proportional weight: $\mu_k = \dfrac{J_k^{\text{cvx}}}{J_k^{\text{cvx}} + J_k^{\text{ps}}}$ if $J_k^{\text{cvx}} + J_k^{\text{ps}} > 0$, and $\mu_k = 1/2$ otherwise.	Không	Giữ nguyên
[98] Combine the two directions: $y^k = x^k - \left[\mu_k \eta_k^{\text{cvx}} g_k^{\text{cvx}} + (1 - \mu_k)\eta_k^{\text{ps}} g_k^{\text{ps}}\right]$.	Không	Giữ nguyên
[99] Step 2 (projection).	Không	Giữ nguyên
[100] $x^{k+1} = P_K(y^k)$; set $k \leftarrow k + 1$ and go to Step 1.	Không	Giữ nguyên
[101] The proportional weight is an adaptive allocation based on the two maximal violations; we do not claim that it is the only choice for which DTAP converges, or that it is optimal.	Không	Giữ nguyên
[102] Its specific form is used in (27) to control the convex-track residual and in Claim 2 of Stage 2 of Theorem 3.12 to ensure that a persistently violated pseudoconvex track retains positive weight.	**Có** — E [3]	33 từ, chuỗi ”in Claim 2 of Stage 2 of Theorem 3.12” nặng. Có thể viết: ”Its specific form enters the proof twice: in (27), to bound the convex-track residual, and in Claim 2 of Theorem 3.12, to keep a persistently violated pseudoconvex track at positive weight.”
[103] Although the one-step inequality of Lemma 3.8 holds for every $\mu_k \in [0, 1]$, it does not by itself imply convergence for every weight sequence.	Không	Giữ nguyên
[104] The iteration does not require prescribed operator norms, Lipschitz constants, or Hölder moduli.	Không	Giữ nguyên
[105] The proof nevertheless uses finite bounds on function values and gradients on a compact set containing the iterates, after boundedness has been proved.	Không	Giữ nguyên
[106] Bounded gradients imply Lipschitz continuity of the functions on that convex compact set, but these bounds are not algorithm inputs.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[107] Assumption (A2) does not imply Lipschitz continuity of the gradients, and the theorem does not require it.	Không	Giữ nguyên
[108] The user still chooses ρ_{cvx}, δ, and $\{\gamma_k\}$; these parameters do not depend on any constant of the problem.	Không	Giữ nguyên
[109] For the analysis we write the combined increment as [công thức (16)] so $y^k = x^k - d_k$ and $x^{k+1} = P_K(x^k - d_k)$.	Không	Giữ nguyên
[110] **Per-iteration cost**	Không	Giữ nguyên
[111] Each DTAP iteration evaluates the componentwise violations $v_i(x^k)$ and the active-set gradients $\{\nabla F_i(x^k) : i \in K_k^{\text{cvx}} \cup K_k^{\text{ps}}\}$, forms the two search directions $g_k^{\text{cvx}}, g_k^{\text{ps}}$ as weighted sums of these gradients, computes the step sizes, and performs one projection onto K.	Không	Giữ nguyên
[112] It uses neither an operator-norm estimate nor a line search.	Không	Giữ nguyên
[113] The cost is dominated by the constraint evaluations and the projection.	Không	Giữ nguyên
[114] The cost for the feature-selection instance is given in Section 4.1 (Remark 6).	Không	Giữ nguyên
[115] **Auxiliary lemmas**	Không	Giữ nguyên
[116] The two tracks need different arguments.	Không	Giữ nguyên
[117] On the convex track, Lemma 3.4 bounds $\langle \nabla F_i(x),\, x - x^\star \rangle$ from below by the violation $v_i(x)$; this gives a decrease of $\left\|x^k - x^\star\right\|^2$ of order $\mu_k T_k$.	**Có** — B [2]	Dùng T_k và x* trước khi định nghĩa (T_k ở (19), Mục 3.7). Viết: ”On the convex track, Lemma 3.4 bounds ⟨∇F_i(x),x−x*⟩, for x*∈Ω, from below by the violation v_i(x); this yields a guaranteed decrease of ‖x^k−x*‖² (Lemma 3.8).”
[118] On the pseudoconvex track, definition (5) gives only the sign of this inner product.	Không	Giữ nguyên
[119] Lemma 3.8 combines both tracks in one quasi-Fejér inequality; the choice of the parameter ζ in Young’s inequality keeps the convex-track term negative.	Không	Giữ nguyên
[120] Stage 2 of Theorem 3.12 then shows, by contradiction, that at least one cluster point is feasible, and Stage 3 shows that the whole sequence converges strongly to it.	Không	Giữ nguyên
[121] The two degenerate cases follow as corollaries.	Không	Giữ nguyên
[122] Throughout, $\{x^k\}$ is the infinite sequence generated by Algorithm 1 with the stopping test removed.	Không	Giữ nguyên
[123] As shown below (Lemma 3.10 and Theorem 3.12), $J(x^k) \to 0$; hence for any $\varepsilon > 0$ there is a finite k with $J(x^k) \le \varepsilon$, so the finite-stopping version terminates after finitely many steps with an ε-feasible point.	**Có** — B [3]	33 từ; ”finite-stopping version” là cách gọi tự đặt. Viết: ”…so Algorithm 1 with its stopping test terminates after finitely many steps with an ε-feasible point.”
[124] We begin with two preparatory results used repeatedly below.	Không	Giữ nguyên
[125] Let $K \subseteq H$ be nonempty closed convex.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[126] For every $y \in H$ and $z \in K$, [công thức (17)]	Không	Giữ nguyên
[127] Put $p := P_K(y)$.	Không	Giữ nguyên
[128] By Proposition 2.1, $\langle y-p,\, z-p\rangle \le 0$, so expanding $\|y-z\|^2 = \|y-p\|^2 + 2\langle y-p,\, p-z\rangle + \|p-z\|^2$ and using $\langle y-p,\, p-z\rangle \ge 0$ gives $\|y-z\|^2 \ge \|y-p\|^2 + \|p-z\|^2$, which is (17).	Không	Giữ nguyên
[129] Let $\{a_k\} \subset [0,\infty)$ and $\{\varepsilon_k\} \subset [0,\infty)$ satisfy $a_{k+1} \le a_k + \varepsilon_k$ for all k and $\sum_{k=0}^{\infty} \varepsilon_k < \infty$.	Không	Giữ nguyên
[130] Then $\{a_k\}$ converges to a finite limit.	Không	Giữ nguyên
[131] $b_k := a_k - \sum_{l<k} \varepsilon_l$ is nonincreasing (from $a_{k+1} \le a_k + \varepsilon_k$) and bounded below by $-\sum_l \varepsilon_l > -\infty$, hence converges; since $\sum_{l<k} \varepsilon_l \to S < \infty$, $a_k = b_k + \sum_{l<k} \varepsilon_l$ converges.	Không	Giữ nguyên
[132] On the convex track, convexity gives a quantitative link between the gradient and the violation.	Không	Giữ nguyên
[133] For $x^\star \in \Omega$, $i \in I^{\mathrm{cvx}}$, and $x \in K$, [công thức]	Không	Giữ nguyên
[134] If $v_i(x) = 0$ both sides vanish.	Không	Giữ nguyên
[135] Let $v_i(x) > 0$, i.e. $F_i(x) > b_i$.	Không	Giữ nguyên
[136] Define $\varphi : [0,1] \to \mathbb{R}$ by $\varphi(t) := F_i((1-t)x + tx^\star)$; since K is convex, $[x, x^\star] \subset K$, so φ is well defined, and by (A1)(b) it is convex, while by (A2) it is differentiable with $\varphi'(0^+) = \langle \nabla F_i(x),\, x^\star - x\rangle$.	Không	Giữ nguyên
[137] The one-dimensional gradient inequality gives $\varphi(1) \ge \varphi(0) + \varphi'(0^+)$, i.e. $F_i(x^\star) \ge F_i(x) + \langle \nabla F_i(x),\, x^\star - x\rangle$.	Không	Giữ nguyên
[138] As $x^\star \in \Omega$ we have $F_i(x^\star) \le b_i < F_i(x)$, whence $\langle \nabla F_i(x),\, x - x^\star\rangle \ge F_i(x) - b_i = v_i(x)$.	Không	Giữ nguyên
[139] Multiplying by $v_i(x) > 0$ gives the claim.	Không	Giữ nguyên
[140] Summing Lemma 3.4 over the approximate active set K_k^{cvx} yields $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \ge \sum_{i \in K_k^{\mathrm{cvx}}} v_i(x^k)^2 = 2\Phi_k^{\mathrm{cvx}}$, a bound used in the proofs of Lemmas 3.8 and 3.10.	Không	Giữ nguyên
[141] On the pseudoconvex track, only a sign is available.	Không	Giữ nguyên
[142] For $x^\star \in \Omega$, $j \in I^{\mathrm{ps}}$, and $x \in K$, $v_j(x)\ \langle \nabla F_j(x),\, x - x^\star\rangle \ge 0$, with strict inequality when $v_j(x) > 0$.	Không	Giữ nguyên
[143] If $v_j(x) = 0$ both sides vanish.	Không	Giữ nguyên
[144] If $v_j(x) > 0$ then $F_j(x) > b_j \ge F_j(x^\star)$.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[145] By (A1)(b), $F_j\|_K$ is pseudoconvex; applying (5) with $y = x^\star$ gives $\langle \nabla F_j(x),\, x^\star - x\rangle < 0$, i.e. $\langle \nabla F_j(x),\, x - x^\star\rangle > 0$.	Không	Giữ nguyên
[146] Multiplying by $v_j(x) > 0$ yields the strict inequality.	Không	Giữ nguyên
[147] Finally, two compactness-based norm bounds on the search directions.	**Có** — E [3]	Câu không có động từ. Viết: ”Finally, we bound the norms of the two search directions on compact sets.”
[148] Assume $I^{\mathrm{cvx}} \neq \emptyset$ and let $B \subset U$ be compact.	Không	Giữ nguyên
[149] With $L_B := \max_{i\in I^{\mathrm{cvx}}} \sup_{x\in B} \|\nabla F_i(x)\|$ (finite by (A2)), for every $x \in B$, [công thức]	Không	Giữ nguyên
[150] Put $K' = K^{\mathrm{cvx}}(x)$, $m' = \|K'\| \le \|I^{\mathrm{cvx}}\|$.	Không	Giữ nguyên
[151] By the triangle and Cauchy–Schwarz inequalities, [công thức]	Không	Giữ nguyên
[152] Since $\sum_{i\in K'} v_i(x)^2 = 2\Phi^{\mathrm{cvx}}(x)$ and $\sum_{i\in K'} \|\nabla F_i(x)\|^2 \le m' L_B^2 \le \|I^{\mathrm{cvx}}\|\, L_B^2$, squaring gives the claim.	Không	Giữ nguyên
[153] Assume $I^{\mathrm{ps}} \neq \emptyset$ and let $B \subset U$ be compact.	Không	Giữ nguyên
[154] With $L'_B := \max_{j\in I^{\mathrm{ps}}} \sup_{x\in B} \|\nabla F_j(x)\|$ and $V_B := \max_{j\in I^{\mathrm{ps}}} \sup_{x\in B} v_j(x)$ (both finite by (A2)), for every $x \in B$, [công thức]	Không	Giữ nguyên
[155] By the triangle inequality, $\|g^{\mathrm{ps}}(x)\| = \big\|\sum_{j\in K^{\mathrm{ps}}(x)} v_j(x)\nabla F_j(x)\big\| \le \sum_{j\in K^{\mathrm{ps}}(x)} v_j(x)\,\|\nabla F_j(x)\| \le \|K^{\mathrm{ps}}(x)\|\, V_B L'_B \le \|I^{\mathrm{ps}}\|\, V_B L'_B$.	Không	Giữ nguyên
[156] **Boundedness and decay of the iteration error**	Không	Giữ nguyên
[157] The following inequality is of quasi-Fejér type in the sense of Combettes [5].	Không	Giữ nguyên
[158] It yields boundedness of $\{x^k\}$ and decay of the convex-track violation.	Không	Giữ nguyên
[159] Set [công thức (18)] and [công thức (19)]	Không	Giữ nguyên
[160] For fixed $x^\star \in \Omega$ write $R_k^{x^\star} := \langle g_k^{\mathrm{ps}},\, x^k - x^\star\rangle$.	Không	Giữ nguyên
[161] For every $x^\star \in \Omega$ and $k \ge 0$, [công thức (20)]	Không	Giữ nguyên
[162] Apply Lemma 3.2 with $y = y^k$ and $z = x^\star \in K$ and use $y^k = x^k - d_k$ from (16): [công thức (21)]	Không	Giữ nguyên
[163] Cross term.	Không	Giữ nguyên
[164] Expanding d_k, [công thức (22)]	Không	Giữ nguyên
[165] If $\|g_k^{\mathrm{cvx}}\| = 0$ then $\eta_k^{\mathrm{cvx}} = 0$ and the first term vanishes.	Không	Giữ nguyên
[166] If $\|g_k^{\mathrm{cvx}}\| > 0$, Remark 3 gives $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \ge 2\Phi_k^{\mathrm{cvx}}$; multiplying by $\mu_k \eta_k^{\mathrm{cvx}} \ge 0$ and substituting $\eta_k^{\mathrm{cvx}} = 2\rho_{\mathrm{cvx}}\Phi_k^{\mathrm{cvx}} / \|g_k^{\mathrm{cvx}}\|^2$, [công thức (23)]	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[167] Quadratic term.	Không	Giữ nguyên
[168] By Young's inequality $\|u+v\|^2 \le (1+\zeta)\|u\|^2 + (1+\zeta^{-1})\|v\|^2$ with $u = \mu_k \eta_k^{\text{cvx}} g_k^{\text{cvx}}$, $v = (1-\mu_k)\eta_k^{\text{ps}} g_k^{\text{ps}}$, [công thức]	Không	Giữ nguyên
[169] By definition $\eta_k^{\text{ps}} = \gamma_k / \max\{1, \|g_k^{\text{ps}}\|\}$, so $\eta_k^{\text{ps}} \|g_k^{\text{ps}}\| \le \gamma_k$ and hence $(\eta_k^{\text{ps}})^2 \|g_k^{\text{ps}}\|^2 \le \gamma_k^2$; with $(1-\mu_k)^2 \le 1$, [công thức (24)]	Không	Giữ nguyên
[170] Choose $\zeta := (2-\rho_{\text{cvx}})/(2\rho_{\text{cvx}}) > 0$.	Không	Giữ nguyên
[171] Then $(1+\zeta)\rho_{\text{cvx}} = (2+\rho_{\text{cvx}})/2$, so $(1+\zeta)\rho_{\text{cvx}}^2 = \rho_{\text{cvx}}(2+\rho_{\text{cvx}})/2$, and $1+\zeta^{-1} = (2+\rho_{\text{cvx}})/(2-\rho_{\text{cvx}}) = C_0$.	Không	Giữ nguyên
[172] Combining (21), (22), (23) and (24), [công thức]	Không	Giữ nguyên
[173] The bracket equals $\mu_k \rho_{\text{cvx}} \big(-2 + \frac{2+\rho_{\text{cvx}}}{2}\mu_k\big)$.	Không	Giữ nguyên
[174] Since $\rho_{\text{cvx}} < 2$, for all $\mu_k \in [0,1]$ the factor $-2 + \frac{2+\rho_{\text{cvx}}}{2}\mu_k$ is at most $-(2-\rho_{\text{cvx}})/2 < 0$ (its maximum over $[0,1]$ is at $\mu_k = 1$); multiplying by $\mu_k \rho_{\text{cvx}} \ge 0$ gives the bracket $\le -c_0 \mu_k$.	Không	Giữ nguyên
[175] This yields (20).	Không	Giữ nguyên
[176] The sequence $\{x^k\}$ is bounded.	Không	Giữ nguyên
[177] Moreover, for each fixed $x^\star \in \Omega$: (i) $\{\|x^k - x^\star\|^2\}$ converges in $[0,\infty)$; (ii) $\sum_{k=0}^{\infty} \mu_k T_k < \infty$; (iii) $\sum_{k=0}^{\infty} (1-\mu_k)\eta_k^{\text{ps}} R_k^{x^\star} < \infty$.	Không	Giữ nguyên
[178] Fix $x^\star \in \Omega$ and set $a_k := \|x^k - x^\star\|^2$.	Không	Giữ nguyên
[179] By Lemma 3.5, $R_k^{x^\star} \ge 0$, and $\mu_k, T_k, (1-\mu_k)\eta_k^{\text{ps}} \ge 0$, so (20) gives $a_{k+1} \le a_k + C_0 \gamma_k^2$.	Không	Giữ nguyên
[180] Since $\sum_k C_0 \gamma_k^2 < \infty$ by (A3)(iii), Lemma 3.3 yields (i), and boundedness of $\{x^k\}$ follows.	Không	Giữ nguyên
[181] Summing (20) from 0 to $N-1$, [công thức]	Không	Giữ nguyên
[182] The right side is bounded uniformly in N; letting $N \to \infty$ and using $c_0 > 0$ with nonnegative terms gives (ii) and (iii).	Không	Giữ nguyên
[183] $\lim_{k\to\infty} J_k^{\text{cvx}} = 0$.	Không	Giữ nguyên
[184] If $I^{\text{cvx}} = \emptyset$ then $J_k^{\text{cvx}} = 0$ for all k.	Không	Giữ nguyên
[185] Assume $I^{\text{cvx}} \ne \emptyset$, fix $x^\star \in \Omega$, and put $B := \overline{\text{conv}}\{x^k : k \ge 0\}$, compact by Lemma 3.9 and contained in $K \subset U$.	Không	Giữ nguyên
[186] By (A2) and compactness, [công thức]	Không	Giữ nguyên
[187] If $V = 0$ then all $v_i(x^k) = 0$ and $J_k^{\text{cvx}} = 0$; assume $V > 0$.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[188] If $L = 0$ then each F_i, $i \in I^{\text{cvx}}$, is constant on B, so $g_k^{\text{cvx}} = 0$ and, by Remark 3, $0 = \langle g_k^{\text{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\text{cvx}} \geq (J_k^{\text{cvx}})^2$, giving $J_k^{\text{cvx}} = 0$; assume $L > 0$.	Không	Giữ nguyên
[189] Set $m^{\text{cvx}} := \lvert I^{\text{cvx}} \rvert \geq 1$.	Không	Giữ nguyên
[190] Consider a step with $J_k^{\text{cvx}} > 0$ and pick $i^\star \in I^{\text{cvx}}$ with $v_{i^\star}(x^k) = J_k^{\text{cvx}}$.	Không	Giữ nguyên
[191] Since $v_{i^\star}(x^k) \geq J_k^{\text{cvx}} - \delta$, we have $i^\star \in K_k^{\text{cvx}}$, whence [công thức (25)]	Không	Giữ nguyên
[192] Remark 3 gives $\langle g_k^{\text{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\text{cvx}} > 0$, so $g_k^{\text{cvx}} \neq 0$; with Lemma 3.6, $\lVert g_k^{\text{cvx}} \rVert^2 \leq 2m^{\text{cvx}} L^2 \Phi_k^{\text{cvx}}$, and with (25), [công thức (26)]	Không	Giữ nguyên
[193] For a lower bound on μ_k: for all $i \in I$ and $x^k \in B$, $v_i(x^k) \leq \lvert F_i(x^k) - b_i \rvert \leq V$, so $J_k^{\text{cvx}} \leq V$ and $J_k^{\text{ps}} \leq V$; when $J_k^{\text{cvx}} > 0$, [công thức (27)]	Không	Giữ nguyên
[194] Combining (26) and (27), for $J_k^{\text{cvx}} > 0$, $\mu_k T_k \geq (J_k^{\text{cvx}})^3 / (2V m^{\text{cvx}} L^2)$; this also holds (as $0 \geq 0$) when $J_k^{\text{cvx}} = 0$.	Không	Giữ nguyên
[195] By Lemma 3.9(ii), [công thức (28)] so $(J_k^{\text{cvx}})^3 \to 0$ and thus $J_k^{\text{cvx}} \to 0$.	Không	Giữ nguyên
[196] [công thức] in particular, $\lVert x^{k+1} - x^k \rVert \to 0$.	Không	Giữ nguyên
[197] Since $x^k \in K$ and $x^{k+1} = P_K(y^k)$, Lemma 3.2 with $y = y^k$, $z = x^k$ gives $\lVert x^{k+1} - x^k \rVert \leq \lVert y^k - x^k \rVert = \lVert d_k \rVert$.	Không	Giữ nguyên
[198] By (24) and $\mu_k^2 \leq \mu_k$, $\lVert d_k \rVert^2 \leq (1+\zeta)\rho_{\text{cvx}}^2 \mu_k T_k + C_0 \gamma_k^2$.	Không	Giữ nguyên
[199] Summing and using Lemma 3.9(ii) and (A3)(iii), both series are finite.	**Có** — A [3]	”both series” chưa nói là chuỗi nào. Viết: ”Summing, the series Σ_k μ_kT_k and Σ_k γ_k² are finite by Lemma 3.9(ii) and (A3)(iii), and the claim follows.”
[200] **Strong convergence to a feasible point**	Không	Giữ nguyên
[201] We now prove that the whole sequence converges strongly to a feasible point.	Không	Giữ nguyên
[202] The proof has three stages.	Không	Giữ nguyên
[203] Stage 1 uses Lemma 3.10 to show that every cluster point satisfies the convex constraints.	Không	Giữ nguyên
[204] Stage 2 shows, by contradiction and pseudoconvexity, that at least one cluster point is feasible.	Không	Giữ nguyên
[205] Stage 3 uses the convergence of $\lVert x^k - \bar{x} \rVert$ to show that the whole sequence converges to that point.	Không	Giữ nguyên
[206] Under Assumption 1, the sequence $\{x^k\}$ generated by Algorithm 1 converges strongly to a point $x^\infty \in \Omega$.	Không	Giữ nguyên
[207] By Lemma 3.9, $\{x^k\}$ is bounded; let $C := \{\bar{x} \in H : \exists\, k_l \to \infty,\ x^{k_l} \to \bar{x}\}$ be its (nonempty, compact) cluster set.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[208] Since K is closed and $x^k \in K$, $C \subseteq K$.	Không	Giữ nguyên
[209] Stage 1 (convex constraints).	Không	Giữ nguyên
[210] Fix $\bar{x} \in C$ and $x^{k_l} \to \bar{x}$.	Không	Giữ nguyên
[211] By (A2) each F_i is continuous, so $v_i(x^{k_l}) \to v_i(\bar{x})$; for $i \in I^{\text{cvx}}$, $v_i(x^{k_l}) \le J^{\text{cvx}}_{k_l} \to 0$ by Lemma 3.10, so by squeezing $v_i(\bar{x}) = 0$, i.e. $F_i(\bar{x}) \le b_i$.	Không	Giữ nguyên
[212] Hence [công thức (29)]	Không	Giữ nguyên
[213] If $I^{\text{ps}} = \emptyset$, then (29) gives $C \subseteq \Omega$, so $C \cap \Omega = C \neq \emptyset$.	Không	Giữ nguyên
[214] Assume henceforth $I^{\text{ps}} \neq \emptyset$.	Không	Giữ nguyên
[215] Stage 2 (a feasible cluster point).	Không	Giữ nguyên
[216] Suppose, for contradiction, that $C \cap \Omega = \emptyset$.	Không	Giữ nguyên
[217] By (29) and the partition $I = I^{\text{cvx}} \cup I^{\text{ps}}$, for each $\bar{x} \in C$ there is $j \in I^{\text{ps}}$ with $F_j(\bar{x}) > b_j$, i.e. $J^{\text{ps}}(\bar{x}) > 0$.	Không	Giữ nguyên
[218] As a maximum of finitely many continuous functions, J^{ps} is continuous and positive on the compact set C, so $\alpha := \min_{x \in C} J^{\text{ps}}(x) > 0$.	Không	Giữ nguyên
[219] Claim 1 (uniform lower bound for J^{ps}_k).	Không	Giữ nguyên
[220] There exist $\tau_0 > 0$ and k_0 with $J^{\text{ps}}_k \ge \tau_0$ for all $k \ge k_0$.	Không	Giữ nguyên
[221] Indeed, put $\tau_0 := \alpha/2$.	Không	Giữ nguyên
[222] If the claim failed there would be $k_l \to \infty$ with $J^{\text{ps}}_{k_l} < \tau_0$; extracting (same notation) $x^{k_l} \to \tilde{x} \in C$, continuity gives $J^{\text{ps}}(\tilde{x}) \le \tau_0 < \alpha$, contradicting the definition of α.	Không	Giữ nguyên
[223] Claim 2 ($\mu_k \to 0$).	Không	Giữ nguyên
[224] For $k \ge k_0$, $\mu_k = J^{\text{cvx}}_k/(J^{\text{cvx}}_k + J^{\text{ps}}_k) \le J^{\text{cvx}}_k/\tau_0 \to 0$ by Lemma 3.10.	Không	Giữ nguyên
[225] Hence there is $k_1 \ge k_0$ with $1 - \mu_k \ge 1/2$ for $k \ge k_1$.	Không	Giữ nguyên
[226] Claim 3 (uniform lower bound for $R^{x^\star}_k$).	Không	Giữ nguyên
[227] Fix $x^\star \in \Omega$.	Không	Giữ nguyên
[228] There exist $\xi_0 > 0$ and $k_2 \ge k_1$ with $R^{x^\star}_k \ge \xi_0$ for all $k \ge k_2$.	Không	Giữ nguyên
[229] Suppose not: there is $k_l \to \infty$ with $R^{x^\star}_{k_l} \to 0$.	Không	Giữ nguyên
[230] As $\{x^{k_l}\}$ is bounded, extract $x^{k_l} \to \bar{x} \in C$; since I^{ps} has finitely many subsets, extract further so that $K^{\text{ps}}_{k_l} \equiv K^\star$ is constant.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[231] By assumption $J^{\mathrm{ps}}(\bar{x}) > 0$; choose $j^\star \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(\bar{x})$, so $v_{j^\star}(\bar{x}) = J^{\mathrm{ps}}(\bar{x}) > 0$.	Không	Giữ nguyên
[232] By continuity $v_{j^\star}(x^{k_l}) \to v_{j^\star}(\bar{x})$ and $J^{\mathrm{ps}}_{k_l} \to J^{\mathrm{ps}}(\bar{x})$, so $J^{\mathrm{ps}}_{k_l} - v_{j^\star}(x^{k_l}) \to 0$; with $\delta > 0$ from (A3)(i), for large l we get $v_{j^\star}(x^{k_l}) \geq J^{\mathrm{ps}}_{k_l} - \delta$, hence $j^\star \in K^{\mathrm{ps}}_{k_l} = K^\star$.	Không	Giữ nguyên
[233] Each summand of $R^{x^\star}_{k_l} = \sum_{j \in K^\star} v_j(x^{k_l}) \left\langle \nabla F_j(x^{k_l}),\, x^{k_l} - x^\star \right\rangle$ is nonnegative by Lemma 3.5, so keeping only $j^\star$, [công thức]	Không	Giữ nguyên
[234] Letting $l \to \infty$ and using continuity of $v_{j^\star}$ and $\nabla F_{j^\star}$, [công thức]	Không	Giữ nguyên
[235] Since $j^\star \in I^{\mathrm{ps}}$ and $x^\star \in \Omega$, $F_{j^\star}(x^\star) \leq b_{j^\star} < F_{j^\star}(\bar{x})$; definition (5) with $x = \bar{x}$, $y = x^\star$ gives $\langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle > 0$, and with $v_{j^\star}(\bar{x}) > 0$ we get $\Lambda > 0$, contradicting $R^{x^\star}_{k_l} \to 0$.	Không	Giữ nguyên
[236] This proves Claim 3.	Không	Giữ nguyên
[237] Reaching the contradiction.	Không	Giữ nguyên
[238] By Lemma 3.7 on $B := \overline{\mathrm{conv}}\{x^k\}$, $\|g^{\mathrm{ps}}_k\| \leq |I^{\mathrm{ps}}|\, V_B L'_B$ with $L'_B := \max_{j \in I^{\mathrm{ps}}} \sup_B \|\nabla F_j\| < \infty$.	Không	Giữ nguyên
[239] Put $G' := \max\{1, |I^{\mathrm{ps}}|\, V_B L'_B\}$; then $\max\{1, \|g^{\mathrm{ps}}_k\|\} \leq G'$ for all k, so $\eta^{\mathrm{ps}}_k = \gamma_k / \max\{1, \|g^{\mathrm{ps}}_k\|\} \geq \gamma_k / G'$.	Không	Giữ nguyên
[240] For $k \geq k_2$, combining with $1 - \mu_k \geq 1/2$ and $R^{x^\star}_k \geq \xi_0$, [công thức] by (A3)(iii), contradicting Lemma 3.9(iii).	Không	Giữ nguyên
[241] Hence $C \cap \Omega \neq \emptyset$.	Không	Giữ nguyên
[242] Stage 3 (whole-sequence convergence).	Không	Giữ nguyên
[243] Fix $\bar{x} \in C \cap \Omega$.	Không	Giữ nguyên
[244] Lemma 3.9(i), applied to $x^\star = \bar{x}$, ensures that $\lim_{k \to \infty} \left\|x^k - \bar{x}\right\|^2$ exists.	Không	Giữ nguyên
[245] Since $\bar{x} \in C$, a subsequence x^{k_l} converges to $\bar{x}$.	Không	Giữ nguyên
[246] Therefore [công thức]	Không	Giữ nguyên
[247] Hence the whole sequence converges to $\bar{x}$, and we set $x^\infty := \bar{x} \in \Omega$.	Không	Giữ nguyên
[248] The argument does not require Ω to be a singleton: one feasible cluster point and the convergence of its distance suffice.	Không	Giữ nguyên
[249] The parameter $\delta > 0$ is not necessary for convergence.	Không	Giữ nguyên
[250] When $\delta = 0$ one argues by subsequence: along $x^{k_l} \to \bar{x} \in C$, choose $j_l \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(x^{k_l})$; since I^{ps} is finite, extract $j_l \equiv j^\star$, and $v_{j^\star}(x^{k_l}) = J^{\mathrm{ps}}(x^{k_l}) \to J^{\mathrm{ps}}(\bar{x}) > 0$, so $j^\star$ attains the maximal violation at $\bar{x}$ and the pseudoconvexity argument goes through.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[251] The role of $\delta > 0$ is to simplify: a maximizer at $\bar{x}$ automatically lies in $K_{k_l}^{\mathrm{ps}}$ for large l, avoiding a 'finitely many indices' extraction.	Không	Giữ nguyên
[252] Larger δ enlarges K_k^{cvx} and K_k^{ps}, which may improve numerical stability when several $v_i(x^k)$ are nearly equal.	Không	Giữ nguyên
[253] From (28), $\sum_k (J_k^{\mathrm{cvx}})^3 < \infty$, whence $\min_{0 \le k < N} J_k^{\mathrm{cvx}} = O(N^{-1/3})$.	Không	Giữ nguyên
[254] This bounds the best convex-track violation over the first N iterates.	Không	Giữ nguyên
[255] It is not a last-iterate bound, a rate for the full residual $J(x^k)$, or a rate for the distance to a feasible point.	Không	Giữ nguyên
[256] For these latter quantities, under (A1)–(A3) alone we assert only qualitative convergence.	Không	Giữ nguyên
[257] **Degenerate constraint structures**	Không	Giữ nguyên
[258] When the constraint structure degenerates (only a convex track or only a pseudoconvex track), DTAP and its analysis simplify, yielding two corollaries of Theorem 3.12.	Không	Giữ nguyên
[259] If $I^{\mathrm{ps}} = \emptyset$, then $J_k^{\mathrm{ps}} \equiv 0$, $g_k^{\mathrm{ps}} \equiv 0$, and $\mu_k \equiv 1$ whenever $J_k^{\mathrm{cvx}} > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}})$.	Không	Giữ nguyên
[260] The sequence converges strongly to a point $x^\infty \in \Omega = \{x \in K : F_i(x) \le b_i \text{ for all } i \in I^{\mathrm{cvx}}\}$.	Không	Giữ nguyên
[261] Here $d_k = \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}}$, and (21) directly gives $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2\eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}}, x^k - x^\star \right\rangle + (\eta_k^{\mathrm{cvx}})^2 \left\|g_k^{\mathrm{cvx}}\right\|^2$.	Không	Giữ nguyên
[262] If $\|g_k^{\mathrm{cvx}}\| = 0$ then $\eta_k^{\mathrm{cvx}} = 0$, $T_k = 0$, and the bound is $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2$.	Không	Giữ nguyên
[263] Otherwise Lemma 3.4 gives $\left\langle g_k^{\mathrm{cvx}}, x^k - x^\star \right\rangle \ge 2\Phi_k^{\mathrm{cvx}}$, and substituting $\eta_k^{\mathrm{cvx}} = 2\rho_{\mathrm{cvx}} \Phi_k^{\mathrm{cvx}} / \left\|g_k^{\mathrm{cvx}}\right\|^2$ together with $(\eta_k^{\mathrm{cvx}})^2 \left\|g_k^{\mathrm{cvx}}\right\|^2 = \eta_k^{\mathrm{cvx}} \cdot 2\rho_{\mathrm{cvx}} \Phi_k^{\mathrm{cvx}}$ yields, after simplification, $2\eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}}, x^k - x^\star \right\rangle - (\eta_k^{\mathrm{cvx}})^2 \left\|g_k^{\mathrm{cvx}}\right\|^2 \ge \rho_{\mathrm{cvx}}(2 - \rho_{\mathrm{cvx}})(2\Phi_k^{\mathrm{cvx}})^2 / \left\|g_k^{\mathrm{cvx}}\right\|^2 = 2c_0 T_k$.	Không	Giữ nguyên
[264] Hence [công thức (30)] valid for all k.	Không	Giữ nguyên
[265] Thus $\{\left\|x^k - x^\star\right\|^2\}$ is nonincreasing and convergent, and summing gives $\sum_k T_k < \infty$.	Không	Giữ nguyên
[266] As in Lemma 3.10, when $J_k^{\mathrm{cvx}} > 0$ one has $T_k \ge (J_k^{\mathrm{cvx}})^2 / (m^{\mathrm{cvx}} L^2)$ (note the exponent 2 here, versus 3 in (28), since no μ_k factor is present), so $\sum_k (J_k^{\mathrm{cvx}})^2 < \infty$ and $J_k^{\mathrm{cvx}} \to 0$.	Không	Giữ nguyên
[267] Finally, $I^{\mathrm{ps}} = \emptyset$ makes Stage 2 of Theorem 3.12 vacuous (indeed $C \subseteq \Omega$ already from (29)), and Stage 3 gives $x^k \to x^\infty \in \Omega$.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[268] If $I^{\text{cvx}} = \emptyset$, then $J_k^{\text{cvx}} \equiv 0$, $g_k^{\text{cvx}} \equiv 0$, and $\mu_k \equiv 0$ whenever $J_k^{\text{ps}} > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta_k^{\text{ps}} g_k^{\text{ps}})$ with $\eta_k^{\text{ps}} = \gamma_k / \max\{1, \|g_k^{\text{ps}}\|\}$.	Không	Giữ nguyên
[269] The sequence converges strongly to a point $x^\infty \in \Omega$.	Không	Giữ nguyên
[270] Now $d_k = \eta_k^{\text{ps}} g_k^{\text{ps}}$ with $\|d_k\|^2 \le \gamma_k^2$, so (21) gives $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2\eta_k^{\text{ps}} R_k^{x^\star} + \gamma_k^2$, which is (20) in this degenerate case ($T_k = 0$, $1 - \mu_k = 1$).	Không	Giữ nguyên
[271] Since $R_k^{x^\star} \ge 0$ (Lemma 3.5), $a_{k+1} \le a_k + \gamma_k^2$, and Lemma 3.3 gives convergence of $\{\left\|x^k - x^\star\right\|^2\}$ and boundedness; summation gives $\sum_k \eta_k^{\text{ps}} R_k^{x^\star} < \infty$.	Không	Giữ nguyên
[272] Stage 1 of Theorem 3.12 gives $C \subseteq K$ (the convex part being vacuous); Stage 2 with $\mu_k = 0$ (so $1 - \mu_k = 1$) produces the same contradiction $\sum \eta_k^{\text{ps}} R_k^{x^\star} = +\infty$, so $C \cap \Omega \neq \emptyset$; Stage 3 gives $x^k \to x^\infty \in \Omega$.	Không	Giữ nguyên
[273] **Numerical experiments**	Không	Giữ nguyên
[274] We separate predictive performance from the numerical solution of the feasibility system.	**Có** — B [2]	”separate” không rõ là tách theo nghĩa nào. Viết: ”We report predictive performance (Sections 4.1–4.3) separately from the numerical solution of the feasibility problems (Section 4.4).”
[275] The classification comparison uses eight datasets.	Không	Giữ nguyên
[276] The solver diagnostic additionally includes Movement Libras and controlled nonconvex pseudoconvex instances.	**Có** — B [2]	”solver diagnostic” là tên không xuất hiện ở tiêu đề mục nào (Mục 4.4 tên là ”Constraint activity and a tuned subgradient comparison”). Viết: ”Section 4.4 additionally uses Movement Libras and controlled nonconvex pseudoconvex instances.”
[277] **Feature-selection model**	Không	Giữ nguyên
[278] We apply the normal-box NSFP to feature selection by combining criteria from the fractional-programming model of Wang et al. [15] and the fuzzy-rough model AFRSFCV [16].	Không	Giữ nguyên
[279] Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be the training samples, $A = \{a_1, \dots, a_p\}$ the features, $X \in \mathbb{R}^{n \times p}$ the data matrix, and $y \in \{1, \dots, r\}^n$ the class labels.	Không	Giữ nguyên
[280] Here n is the number of samples; subscripts index features and superscripts index iterations.	Không	Giữ nguyên
[281] The feature weights belong to the compact convex simplex $K = \Delta_{p-1} = \{w \in \mathbb{R}^p : w \ge 0,\ \mathbf{1}_p^\top w = 1\}$.	Không	Giữ nguyên
[282] All supervised quantities below are estimated from the training data.	Không	Giữ nguyên
[283] **Fractional criterion and N-Fscore**	Không	Giữ nguyên
[284] The supervised similarity matrix $S \in \mathbb{R}^{p \times p}$ has entries [công thức (31)] where Z_i is the discretized feature a_i, C is the class variable, and $I(Z_i; Z_j; C) = I(Z_i; C) + I(Z_j; C) - I((Z_i, Z_j); C)$ is their interaction information.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[285] Entropy and mutual information are estimated from equal-width histograms with ten bins per feature [2].	Không	Giữ nguyên
[286] We set $s_{ij} = 0$ if the denominator is zero.	Không	Giữ nguyên
[287] The redundancy matrix is [công thức (32)] so $Q_R \succeq 0$.	Không	Giữ nguyên
[288] The implementation uses $\delta_Q = \max\{0, -\lambda_{\min}(S)\} + 10^{-10}$.	Không	Giữ nguyên
[289] Let n_t, $\bar{a}_{kt}$, and σ_{kt}^2 be the class size, mean, and population variance of feature a_k in class t, and let $\bar{a}_k$ be its overall mean.	Không	Giữ nguyên
[290] The Fisher scores are [công thức (33)]	Không	Giữ nguyên
[291] These scores are nonnegative.	Không	Giữ nguyên
[292] The denominator regularization prevents division by zero but does not ensure $\rho_k > 0$, since the numerator can vanish.	Không	Giữ nguyên
[293] The fractional program underlying N-Fscore is [công thức (34)] assuming the denominator is positive on K.	Không	Giữ nguyên
[294] The numerator measures redundancy and the denominator relevance.	Không	Giữ nguyên
[295] Our N-Fscore baseline uses the projection-neural-network stage of [15], with the Fisher scores in (33).	Không	Giữ nguyên
[296] It then applies the common selection rule (44), instead of the source method's k-winners-take-all stage.	Không	Giữ nguyên
[297] Its dynamics and explicit-Euler implementation are given in Section 4.2.1.	Không	Giữ nguyên
[298] **Fuzzy-rough criterion and AFRSFCV**	Không	Giữ nguyên
[299] For each class, write $D_t = \{x_i : y_i = t\}$ and $D_t(x_j) = \mathbf{1}_{\{y_j = t\}}$.	Không	Giữ nguyên
[300] AFRSFCV incorporates class variance into a fuzzy relation.	Không	Giữ nguyên
[301] With $a_k(x_i) = X_{ik}$, the implemented relation is [công thức (35)]	Không	Giữ nguyên
[302] The variance floor makes the formula defined when a feature is constant within a class.	Không	Giữ nguyên
[303] The fuzzy lower approximation is [công thức (36)]	Không	Giữ nguyên
[304] Its values form the matrix $\underline{R_k}(D) \in [0,1]^{n \times r}$.	Không	Giữ nguyên
[305] Let $Y_{it} = \mathbf{1}_{\{y_i = t\}}$ be the label matrix and define the decision matrix by [công thức (37)]	Không	Giữ nguyên
[306] Ties are resolved by the smallest class index.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[307] Thus $P_{ik} = 1$ precisely when feature a_k predicts sample x_i correctly under this rule.	Không	Giữ nguyên
[308] The AFRSFCV baseline [16] minimizes [công thức (38)] where $w_k(\tau) = e^{\tau_k} / \sum_{l=1}^{p} e^{\tau_l}$ ensures $w(\tau) \in K$.	Không	Giữ nguyên
[309] The three terms are the label-approximation error, the weight of misclassifying features per sample, and a Tikhonov term.	**Có** — B [3]	”the weight of misclassifying features per sample” hơi nén. Viết rõ: ”...the squared total weight, per sample, of the features that misclassify it, and a Tikhonov term.”
[310] The baseline uses gradient descent for a fixed iteration budget, followed by (44).	Không	Giữ nguyên
[311] The gradient and update are given in Section 4.2.1.	Không	Giữ nguyên
[312] **Feasibility and feature selection**	Không	Giữ nguyên
[313] We impose separate bounds on three criteria, [công thức (39, 40, 41)]	Không	Giữ nguyên
[314] Here F_1, F_2 are the first two terms of (38), without their coefficients, and F_3 is the objective in (34).	Không	Giữ nguyên
[315] For thresholds $b_i > 0$, the resulting problem is [công thức (42)]	Không	Giữ nguyên
[316] This formulation asks for simultaneous acceptability of the three criteria.	**Có** — B [3]	”acceptability” mơ hồ. Viết: ”A feasible w satisfies the three bounds simultaneously.”
[317] It omits the Tikhonov term and does not preserve either source optimization problem or its minimizers.	Không	Giữ nguyên
[318] If $Q_R \succeq 0$ and $\min_k \rho_k > 0$, all three functions in (39)–(41) are smooth and convex on an open convex neighbourhood of K.	Không	Giữ nguyên
[319] Consequently, (42) is an all-convex normal-box NSFP.	Không	Giữ nguyên
[320] Hence (42) satisfies (A1) and (A2) of Assumption 1, with $U = U_\rho$, whenever $\Omega \neq \varnothing$.	Không	Giữ nguyên
[321] The first two functions are squared norms of affine maps.	Không	Giữ nguyên
[322] For $w \in U_\rho$ and any direction h, put $t = \rho^\top w$ and $a = \rho^\top h$.	Không	Giữ nguyên
[323] Direct differentiation gives [công thức (43)]	Không	Giữ nguyên
[324] Thus F_3 is convex and smooth on the open half-space U_ρ.	Không	Giữ nguyên
[325] Since $\rho^\top w \geq \min_k \rho_k > 0$ for $w \in K$, this half-space contains K.	Không	Giữ nguyên
[326] The experiments use $I^{\text{cvx}} = \{1, 2\}$, $I^{\text{ps}} = \{3\}$, which is admissible since convex functions are pseudoconvex; DTAP then uses only sign information for F_3, so no genuinely nonconvex constraint is tested.	Không	Giữ nguyên
[327] The all-convex assignment $I^{\text{cvx}} = \{1, 2, 3\}$ is also admissible, and Theorem 3.12 applies to both.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[328] Proposition 4.1 needs $\min_k \rho_k > 0$; on all instances used here the smallest score is 4.77×10^{-5}.	Không	Giữ nguyên
[329] Hence the safeguard $\max\{\rho^\top w, 10^{-15}\}$ in the implementation is never active, and the implemented F_3 is (41).	Không	Giữ nguyên
[330] The thresholds are calibrated on the training data: b_i is the q_i-th percentile of F_i over $3\,000$ Dirichlet points (seed 42) and the p vertices of K, where a fixed rule chooses q_i from the sample size, the number of features and classes, and the class imbalance (supplementary metadata).	**Có** — E [2]	49 từ. Tách: ”The thresholds are calibrated on the training data. Each b_i is the q_i-th percentile of F_i over 3 000 Dirichlet points (seed 42) and the p vertices of K; a fixed rule chooses q_i from the sample size, the numbers of features and classes, and the class imbalance (supplementary metadata).”
[331] All solvers share these thresholds.	Không	Giữ nguyên
[332] Calibrating each b_i separately does not guarantee $\Omega \neq \varnothing$; Section 4.4 checks feasibility numerically.	Không	Giữ nguyên
[333] Prescribed thresholds would be used as given.	Không	Giữ nguyên
[334] For every method, sort the computed weights as $w^\star_{(1)} \geq \cdots \geq w^\star_{(p)}$ and retain the first s features, where [công thức (44)]	Không	Giữ nguyên
[335] This cumulative-weight rule does not enforce a prescribed feature budget.	Không	Giữ nguyên
[336] Truncating the weights does not preserve the feasibility guarantee for the full vector.	Không	Giữ nguyên
[337] With the precomputed matrices, evaluating $F_1, \nabla F_1$ costs $O(nrp)$, evaluating $F_2, \nabla F_2$ costs $O(np)$, and evaluating $F_3, \nabla F_3$ costs $O(p^2)$.	Không	Giữ nguyên
[338] Projection onto $K = \Delta_{p-1}$ by sorting costs $O(p \log p)$ operations [6].	Không	Giữ nguyên
[339] A dense iteration therefore costs $O(nrp + p^2)$ operations and requires $O(nrp + p^2)$ storage, including Q_R.	Không	Giữ nguyên
[340] Preprocessing and classifier training are separate costs.	Không	Giữ nguyên
[341] **Experimental setup**	Không	Giữ nguyên
[342] The main classification study uses the eight UCI datasets [8] in Table 1.	Không	Giữ nguyên
[343] The DTAP runs reported here start at the uniform weight vector and use $\rho_{\text{cvx}} = 1.98$, $\delta = 10^{-3}$, $\gamma_k = (k + 0.5)^{-0.55}$, tolerance 10^{-6}, and a budget of 1000 iterations.	Không	Giữ nguyên
[344] SG-const uses the step $\nu_k = 10^{-3}$ and SG-dyn the sampled modulus $\widehat{L}$ (Section 4.2.1); tuned moduli are considered only in Section 4.4.	Không	Giữ nguyên
[345] N-Fscore uses $\varepsilon_{\text{PNN}} = 10^{-3}$ and 1000 Euler grid points on $[0, 0.2]$.	Không	Giữ nguyên
[346] AFRSFCV uses learning rate 10^{-3}, $\lambda_1 = 1$, $\lambda_2 = 0.5$, and 1000 steps.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[347] The shared cumulative threshold is $\theta = 0.9$.	Không	Giữ nguyên
[348] [Chú thích bảng] Datasets used for the main classification comparison.	**Có** — B [2]	Câu thứ hai của chú thích dùng ”solver diagnostic” (xem trên). Viết: ”The additional Movement Libras instance is used only in Section 4.4.”
[349] The additional Movement Libras instance is used only in the solver diagnostic.	**Có** — B [2]	Như trên: thay ”in the solver diagnostic” bằng ”in Section 4.4”.
[350] **Baseline solvers**	Không	Giữ nguyên
[351] The following paragraphs specify the baseline solvers: N-Fscore and AFRSFCV for the models of Sections 4.1.1 and 4.1.2, and the subgradient methods SG-const and SG-dyn.	Không	Giữ nguyên
[352] N-Fscore: the projection neural network	Không	Giữ nguyên
[353] To solve the fractional program (34), Wang et al. [15] proposed a neurodynamic method coupling a projection neural network (PNN) with a k-winners-take-all network.	Không	Giữ nguyên
[354] The PNN is the differential system [công thức (45)] with output $w = g(u)$, where $g(u) = \max\{0, u\}$ is the componentwise ReLU, $\Pi = \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top$, ∇f is the gradient of the objective in (34), and $\varepsilon_{\text{PNN}} > 0$ is a time constant.	Không	Giữ nguyên
[355] We integrate (45) by the explicit Euler method on a uniform grid of N_{grid} points in $[0, T]$, without a stopping rule, using the parameters of Section 4.2.	Không	Giữ nguyên
[356] The weight vector is the projection $P_K\big(g(u(T))\big)$ of the final network output onto K.	Không	Giữ nguyên
[357] For a uniform comparison we use only the PNN part to obtain $w^\star$; the k-WTA stage is replaced by the shared cumulative-threshold rule of Section 4.1.3.	Không	Giữ nguyên
[358] AFRSFCV: gradient and update	Không	Giữ nguyên
[359] The softmax loss (38) is minimized over $\tau \in \mathbb{R}^p$ by gradient descent.	Không	Giữ nguyên
[360] Its gradient is [công thức (46)] with $\nabla_w L = -\mathbf{M}^\top(\text{vec}(Y) - \mathbf{M}w) + \lambda_1(P^\top Pw - P^\top\mathbf{1}_n) + \lambda_2 w$, where $\mathbf{M} \in \mathbb{R}^{nr\times p}$ has k-th column $\text{vec}(\underline{R_k}(D))$.	Không	Giữ nguyên
[361] The update $\tau^{t+1} = \tau^t - \eta\,\nabla_\tau L(\tau^t)$ runs for a fixed budget of T steps; the final weight $w^\star = w(\tau^T) \in K$ feeds the shared selection rule of Section 4.1.3.	**Có** — C [3]	Ký hiệu trùng: η là tốc độ học của AFRSFCV nhưng η_k^cvx, η_k^ps là độ dài bước của DTAP; T là số bước ở đây, thời điểm cuối của PNN ở (45), và T_k ở (19). Có thể đổi tốc độ học thành α và số bước thành N_AF.
[362] SG-const and SG-dyn	Không	Giữ nguyên
[363] These subgradient methods for quasiconvex feasibility [7] follow its most-violated-constraint strategy: with $\mathcal{I}_k = \arg\max_i v_i(w^k)$, [công thức]	Không	Giữ nguyên
[364] SG-const uses $\nu_k = 10^{-3}$; SG-dyn uses the dynamic step $\nu_k = J(w^k)/L$ (Hölder exponent 1), with $L = \widehat{L}$, the largest gradient norm at 500 points sampled in K.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[365] **Evaluation protocol**	Không	Giữ nguyên
[366] Each dataset uses a single stratified $80/20$ training-pool/test split (seed 42).	Không	Giữ nguyên
[367] Min–max scaling is fitted once on the full training pool, not separately within its selection folds, and is then applied to the test data.	Không	Giữ nguyên
[368] Stratified folds (seed 42) within the pool use $N_f = 10$, reduced to 7 on Glass and 4 on Yeast.	Không	Giữ nguyên
[369] The fold masks use $\theta = 0.9$ in (44); the fixed strict-majority rule retains features selected in at least $\lfloor N_f/2 \rfloor + 1$ folds.	Không	Giữ nguyên
[370] We denote the voted feature set by $A^\star$.	Không	Giữ nguyên
[371] The final classifiers use the full pool with these voted features: RBF-SVM ($C = 1$, gamma=scale) and 5-NN, with the full feature set as a reference.	Không	Giữ nguyên
[372] Accuracy and macro one-vs-rest AUC are reported.	Không	Giữ nguyên
[373] The voted masks are computed once and then reused: the evaluation stage trains only the downstream classifiers on these fixed masks, with SVM probability-calibration seed 42; identical masks share the same predictions.	Không	Giữ nguyên
[374] The test data are not used to select features or to choose any parameter or vote threshold.	Không	Giữ nguyên
[375] The training/test split is fixed before the selections are computed, and the rule that chooses the threshold levels uses only the sample size, the number of features, the number of classes and the class sizes of the whole dataset, not any feature values.	**Có** — E [2]	43 từ và lặp lại nội dung quy tắc chọn ngưỡng đã nêu ở Mục 4.1.3. Viết: ”The training/test split is fixed before the selections are computed, and the threshold rule of Section 4.1.3 uses only dataset sizes, not feature values.”
[376] Because a single split is used, this is an exploratory evaluation, not an end-to-end or nested-validation study.	Không	Giữ nguyên
[377] For the fold-selection matrix $Z \in \{0,1\}^{N_f \times p}$, the Nogueira index [13] is computed as [công thức]	Không	Giữ nguyên
[378] Here $\bar{Z}_j = N_f^{-1} \sum_b Z_{bj}$, and s_j^2 is the unbiased variance over the N_f folds.	Không	Giữ nguyên
[379] Since the training folds overlap, stability is reported descriptively.	Không	Giữ nguyên
[380] For each classifier metric, we use a two-sided signed-rank test across the eight datasets, omitting zero differences, assigning midranks to ties, and enumerating all signs of the nonzero differences; all-zero pairs receive the p-value 1.	Không	Giữ nguyên
[381] Holm adjustment covers the four DTAP–baseline comparisons separately for each metric.	Không	Giữ nguyên
[382] The Friedman test compares the five selection methods.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[383] **Classification results**	Không	Giữ nguyên
[384] Tables 2–4 report selected-set size, stability, and predictive performance.	Không	Giữ nguyên
[385] DTAP retains 84.2% of the features on average, including $33/36$ on Satimage and $49/57$ on Spambase.	Không	Giữ nguyên
[386] Its mean stability is 0.786, compared with N-Fscore's 0.822.	Không	Giữ nguyên
[387] Thus DTAP reduces the number of features moderately; it does not select sparser or more stable sets than the base models.	Không	Giữ nguyên
[388] AFRSFCV selects much smaller sets on several datasets, with lower mean accuracy under the fixed configuration.	Không	Giữ nguyên
[389] DTAP's mean accuracy is within 0.44 percentage points of the full-feature reference for both classifiers.	Không	Giữ nguyên
[390] It exceeds N-Fscore and AFRSFCV by 3.12 and 4.83 points on SVM accuracy, while retaining more features.	Không	Giữ nguyên
[391] This is a size–prediction tradeoff, not a matched-budget comparison.	Không	Giữ nguyên
[392] Both SG variants have exactly the same voted masks as DTAP on seven datasets; only Spambase differs (48 versus 49 features).	Không	Giữ nguyên
[393] SG-dyn used the full 1000-iteration budget in 6/7 selection folds on Glass, 7/10 on Diabetes and 10/10 on Abalone, and SG-const in 1/10 on Abalone (DTAP in none); the voted weights of both SG variants are the best iterates found within the budget.	**Có** — E [2]	43 từ, nhiều số liệu trong một câu. Tách: "SG-dyn used the full 1000-iteration budget in 6/7 selection folds on Glass, 7/10 on Diabetes and 10/10 on Abalone; SG-const did so in 1/10 folds on Abalone, and DTAP in none. For both SG variants, the voted weights are the best iterates found within the budget."
[394] The SG results on Spambase, the only dataset where they differ from DTAP, are given below Table 3; Table 4 includes all five methods.	Không	Giữ nguyên
[395] Fold-wise stability can differ even when the final voted mask is identical.	Không	Giữ nguyên
[396] Table 5 reports the signed-rank tests.	Không	Giữ nguyên
[397] None of the pairwise comparisons rejects at 0.05 after Holm adjustment; the SVM comparisons with the two base models have adjusted p-values between 0.0625 and 0.0938.	Không	Giữ nguyên
[398] The asymptotic Friedman tests reject equality globally, but this does not establish a particular pairwise superiority.	Không	Giữ nguyên
[399] All DTAP–SG pairwise p-values equal 1; this reflects ties and limited evidence, not a statistical equivalence claim.	Không	Giữ nguyên
[400] These small-sample comparisons concern eight heterogeneous datasets and do not measure variation under repeated train/test splits.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[401] [Chú thích bảng] Selected count $\lvert A^\star \rvert$ / Nogueira stability Ψ; p is the original dimension.	Không	Giữ nguyên
[402] All five selection methods are shown.	Không	Giữ nguyên
[403] [Chú thích bảng] Per-dataset accuracy (%) / macro one-vs-rest AUC on the single holdout.	Không	Giữ nguyên
[404] Equal voted masks reuse the same predictions; the SG exceptions are stated below.	Không	Giữ nguyên
[405] [Chú thích bảng] Unweighted mean classification performance over the eight classification datasets.	Không	Giữ nguyên
[406] All five selection methods and the full-feature reference are included.	Không	Giữ nguyên
[407] [Chú thích bảng] DTAP versus each baseline: raw / Holm-adjusted signed-rank p-values.	Không	Giữ nguyên
[408] Each metric has four comparisons over eight datasets.	Không	Giữ nguyên
[409] The last column gives the tie-corrected Friedman asymptotic p-value for all five methods.	Không	Giữ nguyên
[410] **Constraint activity and a tuned subgradient comparison**	Không	Giữ nguyên
[411] As noted in Section 4.1.3, the calibrated thresholds need not yield $\Omega \neq \varnothing$.	Không	Giữ nguyên
[412] For each full-data instance below, we keep the evaluated point with the largest slack $\min_i(b_i - F_i(w))$ over all runs.	Không	Giữ nguyên
[413] Re-evaluated on matrices rebuilt from the data, each of these points satisfies all three inequalities strictly; the smallest slack is 4.33×10^{-4}.	Không	Giữ nguyên
[414] This is a numerical check, not an exact certificate of feasibility.	Không	Giữ nguyên
[415] We count the iterates of a run, including the last (its evaluated points), at which $F_3(w^k) - b_3 > 10^{-6}$.	**Có** — B [2]	Ngoặc ”(its evaluated points)” khó hiểu, trong khi ”evaluated points” được dùng suốt Mục 4.4. Viết: ”We call the iterates of a run, including the last one, its evaluated points, and count those at which F_3(w^k)−b_3 > 10^{-6}.”
[416] This measures violation along the trajectory, rather than whether the constraint is binding at its limit.	Không	Giữ nguyên
[417] Table 6 shows that, from the standard start, F_3 is violated at no evaluated point on seven of the eight classification datasets, and at one of the 48 evaluated points on Abalone.	Không	Giữ nguyên
[418] Since F_3 is convex (Proposition 4.1), these feature-selection instances test a convex subclass and do not demonstrate a need for the nonconvex extension.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[419] We also report the all-convex comparator (AC), which assigns all three constraints to the convex track.	Không	Giữ nguyên
[420] Protocol and tuning cost.	Không	Giữ nguyên
[421] These are optimization-only tests on full-data feasibility instances.	Không	Giữ nguyên
[422] They use $J \leq 10^{-6}$, at most $1\,000$ evaluated points, and identical starting points across methods.	Không	Giữ nguyên
[423] SG-const and SG-dyn are defined in Section 4.2.1.	Không	Giữ nguyên
[424] Besides SG-dyn with the sampled modulus $\widehat{L}$, a tuned variant selects $L = c\widehat{L}$ from the manually specified grid $c \in \{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\}$ using three random starting points (seeds 1101–1103).	Không	Giữ nguyên
[425] Selection first maximizes the number of successful runs, then minimizes total evaluations, then total best residual.	Không	Giữ nguyên
[426] Evaluation uses the standard start and three different random starts (seeds 2101–2103).	Không	Giữ nguyên
[427] The feature starts are Dirichlet samples.	Không	Giữ nguyên
[428] Only starting points are held out: tuning and evaluation use the same full-data instance, and no classification score is used.	Không	Giữ nguyên
[429] Neither $\widehat{L}$ nor the tuned value is a certified global Hölder modulus.	Không	Giữ nguyên
[430] [Chú thích bảng] Optimization evaluations from the uniform simplex start.	Không	Giữ nguyên
[431] AC denotes the all-convex comparator; sampled and tuned refer to SG-dyn.	Không	Giữ nguyên
[432] † means the tolerance was not reached within the budget.	Không	Giữ nguyên
[433] The last column gives the DTAP third-constraint violation count divided by its number of evaluated points.	Không	Giữ nguyên
[434] Whenever $F_3(w^k) \leq b_3$ along a run, the pseudoconvex track is empty, $\mu_k = 1$ and DTAP coincides step by step with the all-convex comparator; on the eight instances with zero F_3 activity, the identical DTAP and AC columns reflect this identity.	**Có** — E [2]	38 từ. Tách: ”Whenever F_3(w^k) ≤ b_3 along a run, the pseudo-convex track is empty, μ_k = 1, and DTAP coincides step by step with AC. This explains the identical DTAP and AC columns on the eight instances with zero F_3 activity.”
[435] On Abalone, F_3 is violated only at the uniform start ($1/48$), where $J_0^{\mathrm{ps}} \approx 1.56\times10^{-3}$ is negligible against $J_0^{\mathrm{cvx}} \approx 366.3$, so $1-\mu_0 \approx 4.27\times10^{-6}$; the values of J along the two trajectories satisfy $\lvert J_{\mathrm{DTAP}} - J_{\mathrm{AC}}\rvert \leq 3.18\times10^{-5}\, J_{\mathrm{AC}}$ at every evaluated point, and both runs stop after 48 evaluated points.	**Có** — E [2]	41 từ. Tách ba câu: ”On Abalone, F_3 is violated only at the uniform start (1/48), where J^ps_0 ≈ 1.56×10⁻³ is negligible against J^cvx_0 ≈ 366.3, so 1−μ_0 ≈ 4.27×10⁻⁶. Along the two trajectories, |J_DTAP − J_AC| ≤ 3.18×10⁻⁵ J_AC at every evaluated point. Both runs stop after 48 evaluated points.”
[436] The columns are therefore not a comparison of the two assignments, whose success counts differ only on the random starts ($26/36$ versus $36/36$).	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[437] Tuning raises SG-dyn's standard-start success count from $5/9$ to $9/9$.	Không	Giữ nguyên
[438] It can outperform DTAP, for example on Abalone, and SG-const is faster on Movement Libras.	**Có** — A [2]	"It" không rõ là SG-dyn đã tinh chỉnh hay việc tinh chỉnh; "faster" chưa nói theo thước đo nào. Viết: "Tuned SG-dyn can outperform DTAP, for example on Abalone, and SG-const needs fewer evaluations on Movement Libras."
[439] Across all four starts on all nine feature instances, the success counts are $26/36$ for DTAP, $36/36$ for the all-convex comparator, $33/36$ for SG-const, $15/36$ for sampled SG-dyn, and $28/36$ for tuned SG-dyn.	Không	Giữ nguyên
[440] A balanced DTAP variant also obtains $26/36$; it assigns equal weights when both tracks are violated and full weight to the only violated track otherwise.	Không	Giữ nguyên
[441] Thus these tests do not establish a uniform advantage of the proportional weight or robustness to initialization.	Không	Giữ nguyên
[442] The nine feature-instance searches required $179\,744$ optimization evaluations and 10.80 seconds of tuning solver time in total.	Không	Giữ nguyên
[443] These costs are additional to Table 6; DTAP was not tuned.	Không	Giữ nguyên
[444] Initial modulus sampling, preprocessing and search overhead are excluded from this solver-time total.	Không	Giữ nguyên
[445] The supplementary records give the cost of each search separately.	Không	Giữ nguyên
[446] A genuinely nonconvex pseudoconvex constraint.	Không	Giữ nguyên
[447] To exercise the pseudoconvex track directly, consider $K = [-2, 2]^d$ and [công thức (47)]	Không	Giữ nguyên
[448] Here $d \in \{2, 10, 50\}$ and $\kappa \in \{0.3, 1, 3\}$ give nine configurations.	Không	Giữ nguyên
[449] The third function is pseudoconvex: it is an increasing differentiable transform, with positive derivative, of a convex quadratic.	Không	Giữ nguyên
[450] It is not convex, since its radial Hessian eigenvalue is $2\kappa(1 - r^2)/(1 + r^2)^2 < 0$ for $r = \|x - e_1\| > 1$.	Không	Giữ nguyên
[451] The exact witness $x = -e_1/10$ has positive slacks $19/100$, $31/25$, and $\kappa \log(300/221)$.	Không	Giữ nguyên
[452] The standard start is $1.5\mathbf{1}$; the three evaluation starts and three tuning starts are drawn uniformly from $[0.8, 1.8]^d$ with the seeds above (NumPy default_rng).	Không	Giữ nguyên
[453] For modulus sampling we use 500 uniform points in K (seed 42).	Không	Giữ nguyên
[454] SG-const uses the same step $\nu_k = 10^{-3}$ as on the feature instances.	Không	Giữ nguyên

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[455] Tuned SG-const uses a constant step ν selected from the grid $\{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\}$ with the tuning starts and selection rule of tuned SG-dyn; it selects $\nu = 10^{-0.5}$ in all nine configurations.	Không	Giữ nguyên
[456] The balanced variant is the one described above.	Không	Giữ nguyên
[457] [Chú thích bảng] Nonconvex pseudoconvex tests, grouped by κ.	Không	Giữ nguyên
[458] Each entry is successes out of 12 runs, followed by the median number of evaluated points in parentheses, including capped runs.	Không	Giữ nguyên
[459] Each group contains all three dimensions and four starts.	Không	Giữ nguyên
[460] Tuned SG-const uses the step selected as described in the text.	Không	Giữ nguyên
[461] DTAP encounters a third-constraint violation in every one of these 36 runs.	Không	Giữ nguyên
[462] Its pooled activity is $12\,835/12\,877$ evaluated points.	Không	Giữ nguyên
[463] Nevertheless, it reaches the tolerance in only $24/36$ runs, compared with $32/36$ for sampled SG-dyn, $36/36$ for tuned SG-dyn and $36/36$ for tuned SG-const (Table 7).	Không	Giữ nguyên
[464] Tuned SG-const reaches the tolerance in every run with a median of 15–16 evaluated points, fewer than any other method at $\kappa = 0.3$ and $\kappa = 1$; at $\kappa = 3$, DTAP and the balanced variant need a median of 4.	**Có** — E [3]	36 từ. Tách tại dấu chấm phẩy thành hai câu.
[465] These counts exclude the $147\,331$ evaluations of its tuning search.	Không	Giữ nguyên
[466] With $\nu_k = 10^{-3}$, SG-const moves at most $999 \times 10^{-3} = 0.999$ from its start within the budget, whereas every start lies at distance at least 1.11 from Ω; its zero counts are therefore fixed by the step size and the budget.	**Có** — E [3]	37 từ; có thể tách tại dấu chấm phẩy: "…from Ω. Its zero counts are therefore fixed by the step size and the budget."
[467] For $\kappa \leq 1$ the convex track is satisfied from the second evaluated point on, so DTAP then follows the single-track iteration of Corollary 3.14; both tracks are violated simultaneously at only 32 of these $12\,877$ evaluated points.	**Có** — E [3]	36 từ. Tách tại dấu chấm phẩy thành hai câu.
[468] All DTAP failures occur at $\kappa = 0.3$.	Không	Giữ nguyên
[469] Changing κ leaves the feasible set unchanged, but v_3 and ∇F_3 both scale with κ, so when $\|g_k^{\mathrm{ps}}\| \leq 1$ the pseudoconvex step $(1 - \mu_k)\gamma_k g_k^{\mathrm{ps}}$ scales with $\gamma_k \kappa^2$.	Không	Giữ nguyên
[470] Table 8 changes one DTAP parameter at a time for $\kappa \in \{0.3, 1\}$.	Không	Giữ nguyên
[471] With $\gamma_k = a(k + 0.5)^{-0.55}$, $a = 10$ and $a = 100$ give $12/12$ successes at $\kappa = 0.3$; $\rho_{\mathrm{cvx}} = 1$ gives $4/12$, all with $d = 2$; $\delta = 10^{-1}$ reproduces the baseline iterates exactly (Remark 4); the faster-decaying steps $s = 0.75$ and $s = 1$ give no success at $\kappa = 0.3$, and $s = 1$ also fails in all 12 runs at $\kappa = 1$.	**Có** — E [2]	45 từ, bốn kết quả nối bằng dấu chấm phẩy. Tách mỗi kết quả thành một câu, ví dụ: "With γ_k = a(k+0.5)^{-0.55}, a = 10 and a = 100 give 12/12 successes at κ = 0.3. The setting ρ_cvx = 1 gives 4/12, all with d = 2. The setting δ = 10⁻¹ reproduces the baseline iterates exactly (Remark 4). The faster-decaying steps s = 0.75 and s = 1 give no success at κ = 0.3, and s = 1 also fails in all 12 runs at κ = 1."

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[472] The failures at $\kappa = 0.3$ are therefore consistent with a diminishing step that is too small for this constraint scale, rather than with a change in feasibility.	Không	Giữ nguyên
[473] These tests vary one parameter at a time on one problem family and do not identify a setting that is best in general.	Không	Giữ nguyên
[474] On (47), $\delta = 10^{-1}$ and $\delta = 10^{-3}$ give identical iterates because the pseudoconvex track has a single index, at every start either only F_1 is violated or $v_1 - v_2 \geq 4.79$, and the convex track is satisfied from the second evaluated point on; the δ row of Table 8 is therefore a consistency check, not evidence about the role of δ.	**Có** — E [2]	56 từ, câu dài nhất bài. Tách: ”On (47), δ = 10⁻¹ and δ = 10⁻³ give identical iterates. The pseudoconvex track has a single index; at every start, either only F_1 is violated or v_1 − v_2 ≥ 4.79; and the convex track is satisfied from the second evaluated point on. The δ row of Table 8 is therefore a consistency check, not evidence about the role of δ.”
[475] The synthetic tuning searches consumed $188\,295$ optimization evaluations and 10.98 seconds of solver time for SG-dyn, and $147\,331$ evaluations and 8.27 seconds for SG-const, with the same exclusions as above, in the environment stated below.	**Có** — E [3]	35 từ; có thể tách thành hai câu (SG-dyn; SG-const).
[476] [Chú thích bảng] DTAP on (47) with one parameter changed at a time, $\gamma_k = a(k + 0.5)^{-s}$.	Không	Giữ nguyên
[477] The baseline $\rho_{\text{cvx}} = 1.98$, $\delta = 10^{-3}$, $a = 1$, $s = 0.55$ is the DTAP column of Table 7.	Không	Giữ nguyên
[478] Each entry is successes out of 12 runs (three dimensions, four starts), followed by the median number of evaluated points in parentheses, including capped runs.	Không	Giữ nguyên
[479] Reproducibility.	Không	Giữ nguyên
[480] The feature-selection results were saved with Python 3.11.7 and NumPy 1.26.4 on Windows, with one BLAS/OpenMP thread requested.	Không	Giữ nguyên
[481] The feature oracles use algebraically compressed quadratic forms; checked differences from the original values and gradients are at most 1.82×10^{-12} and 1.10×10^{-11}, respectively.	Không	Giữ nguyên
[482] The solver times reported for the feature instances refer to this compressed implementation.	Không	Giữ nguyên
[483] The accompanying package contains only the uncompressed oracles; it reproduces the evaluation counts, while the reported solver times are those of the original compressed implementation.	Không	Giữ nguyên
[484] Tables 7 and 8 and the synthetic tuning costs were produced with Python 3.13.4 and NumPy 2.1.3 on Windows with the default number of BLAS threads; their counts are deterministic, and their solver times are single measurements on that machine.	**Có** — E [3]	40 từ; có thể tách tại dấu chấm phẩy.
[485] With Python 3.13.4 and NumPy 2.1.3, the package also reproduces exactly every numerical entry of Tables 1–5 (Tables 4 and 5 from unrounded scores) and of Table 6, and the counts quoted in Sections 4.3 and 4.4.	**Có** — E [3]	37 từ; chấp nhận được trong đoạn tái lập, có thể giữ.

Câu	Có lỗi không?	Không cần sửa / Cách sửa
[486] The supplementary scripts, raw trajectories, tuning records, witnesses, source hashes and validation instructions accompany the results.	Không	Giữ nguyên
[487] **Conclusion**	Không	Giữ nguyên
[488] DTAP solves normal-box nonlinear split feasibility problems with convex and pseudoconvex components by combining a residual-scaled convex step with a safeguarded pseudoconvex step.	Không	Giữ nguyên
[489] Under consistency and continuous differentiability, the whole sequence converges to a feasible point; no regularity modulus is an input, although the analysis uses bounds on a compact set containing the iterates.	Không	Giữ nguyên
[490] In the feature-selection application all constraints are convex, and test problems with a nonconvex pseudoconvex constraint exercise the pseudoconvex track.	Không	Giữ nguyên
[491] In the experiments, tuned subgradient methods can match or outperform DTAP, whose finite-budget performance depends on the starting point and the step scale.	Không	Giữ nguyên
[492] Rates for the full residual $J(x^k)$, beyond the convex-track bound of Remark 5, and guarantees under error-bound conditions remain open.	Không	Giữ nguyên
[493] **Declaration of competing interest**	Không	Giữ nguyên
[494] The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.	Không	Giữ nguyên
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[496] The authors received no specific funding for this work.	Không	Giữ nguyên
[497] **Declaration of generative AI use**	Không	Giữ nguyên
[498] OpenAI Codex was used during revision for language editing.	Không	Giữ nguyên
[499] **Data availability**	Không	Giữ nguyên
[500] The datasets are available from the UCI Machine Learning Repository [8].	Không	Giữ nguyên
[501] The accompanying reproducibility package contains the scripts, recorded numerical outputs, and instructions for regenerating the reported tables.	Không	Giữ nguyên
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